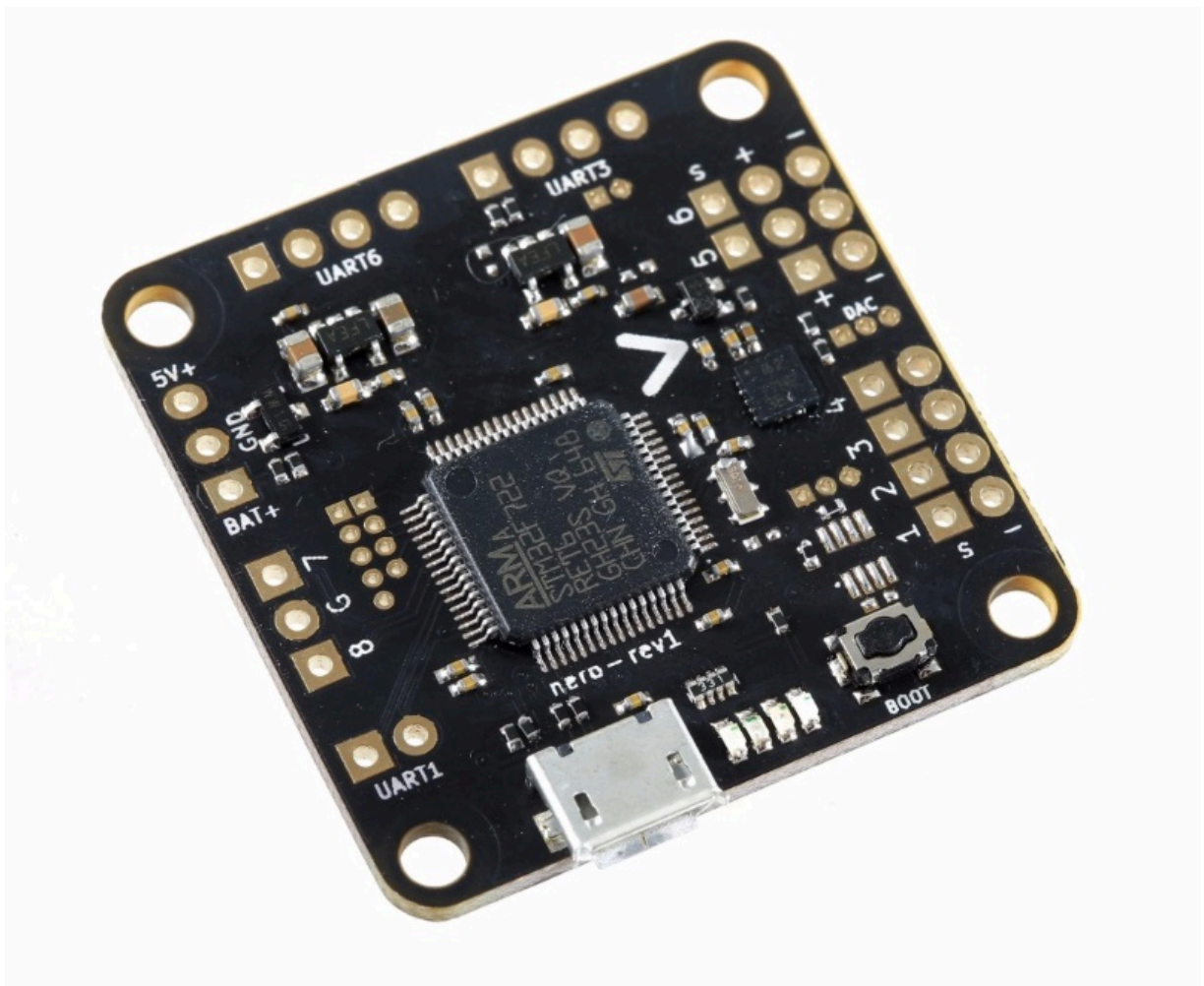


[Boards](#)[Archive](#)[NERO](#)

NERO

Description

Beautifully simple STM32F7 based flightcontroller. A F7 replacement for the Naze.





MCU, Sensors and Features

Hardware

- Size: 36x36mm (30.5x30.5 mounting holes)
- MCU: STM32F722RET6
- IMU: ICM-20602 (SPI)
- IMU Interrupt: Yes
- VCP: Yes
- Hardware UARTS: 3
- OSD: Compatible pin-outs for MinimOSD on UART3 (stackable)
- Blackbox: SD card
- PPM/UART Shared: UART6
- Battery Voltage Sensor: Yes, directly connected, no wiring necessary (if using pololu on full size)
- Integrated Voltage Regulator: Pololu piggy back option

- Button for putting board into DFU mode

Features

- Current Sensor: available as ADC input, but requires shunt circuit on PDB or battery cable.
- BlHeli passthrough: Yes
- WS2811 Led Strip: Yes (on motor output Pin 5)
- Transponder: No
- SPI (2) is broken out for adding an SPI peripheral, e.g. another GYRO.

Manufacturers and Distributors

These boards are now currently available. Shipping of pre-orders is occurring now.

Available here:

- fpvgame.eu
- electricwingman.com
- readytoflyquads.com

More information available here:

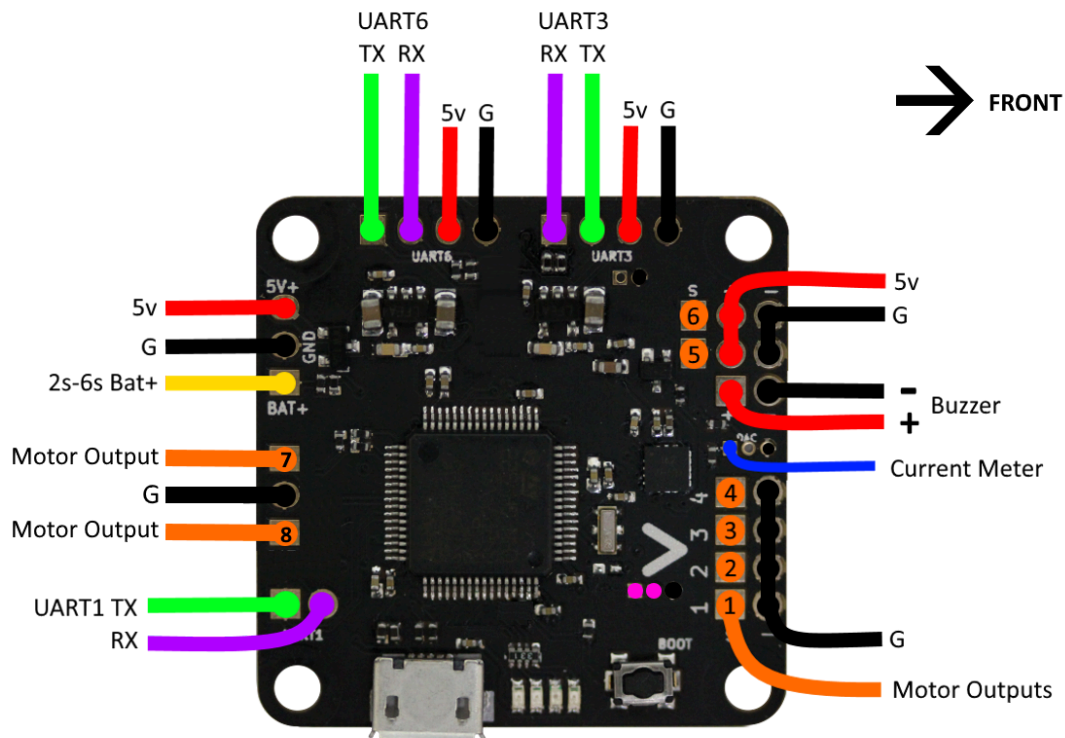
- nerofc.com

Configuration Information

Wiring Diagrams

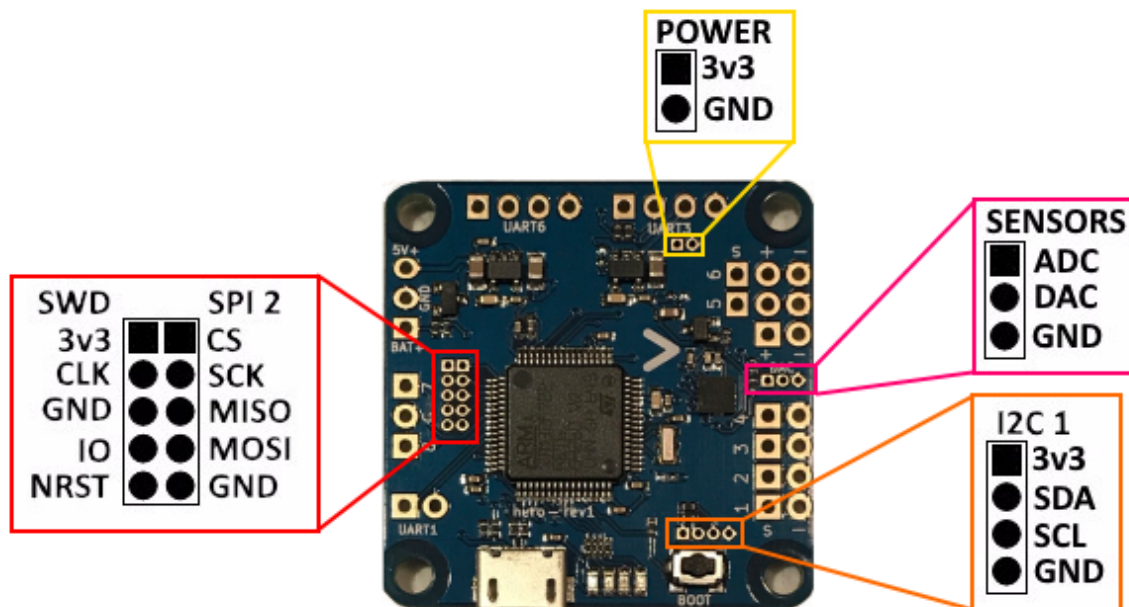
Main wiring details:

NERO - wiring diagram



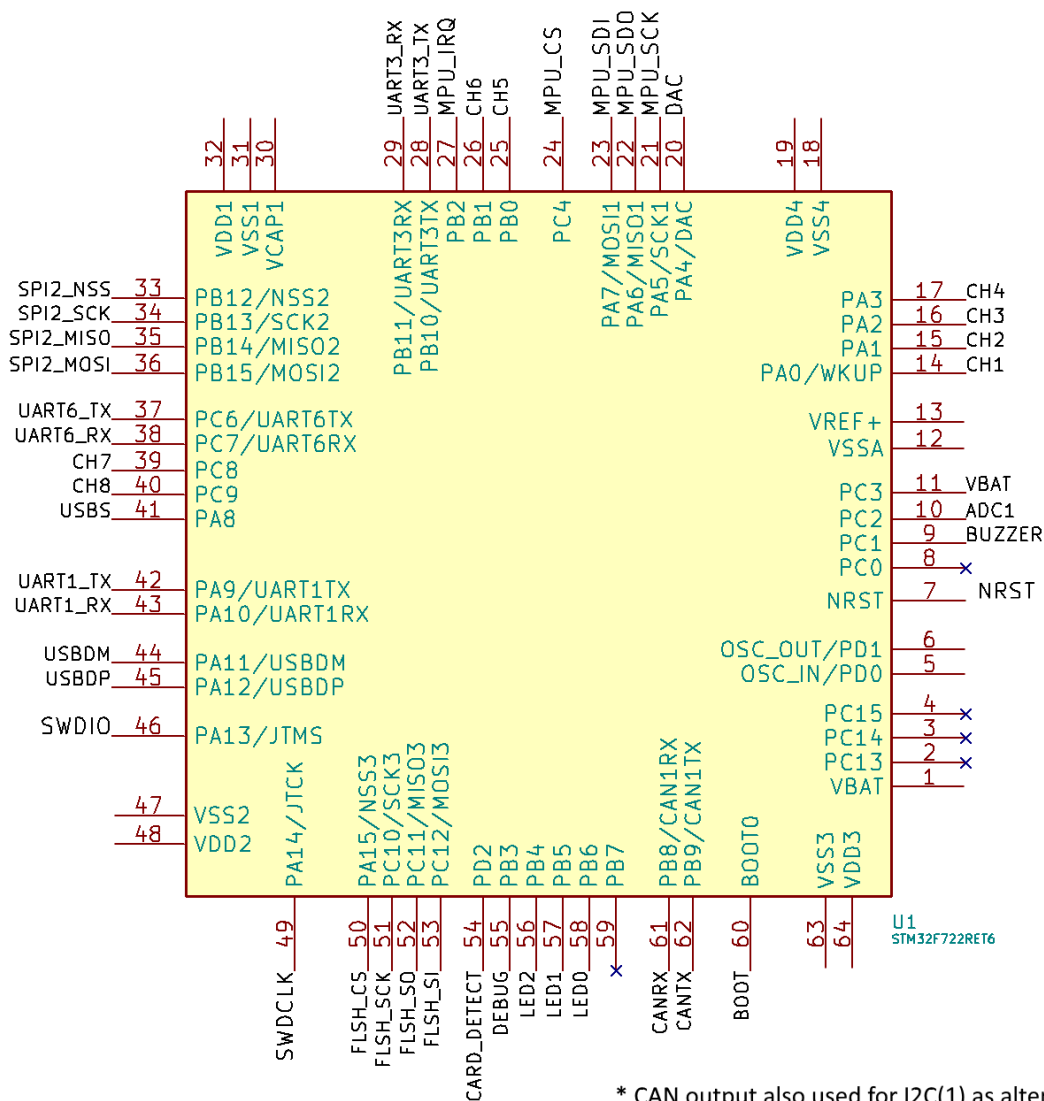
LED Strip output on Motor 5.

Micro pin details:



Schematics

The pin out for the MCU is provided here, so that it can be used as a reference for others considering developing a board using the same target. This is so targets can be minimised going forward. Hopefully the info will also assist other developers in adding features.



* CAN output also used for I2C(1) as alternative

Other Resources

RC Groups Thread: <https://www.rcgroups.com/forums/showthread.php?2734745-NERO-STM32F7-based-FC>

Youtube

Joshua Bardwell

Links

Oscar Liang

 Ivan Efimov

 VitroidFPV

Community

 Discord

 IntoFPV

Feeds

 RSS

 Atom

 JSON



Copyright © 2025 All rights reserved Team Betaflight

Built with Docusaurus

made with ❤️ by **VitroidFPV** and **un!t**